Experiment 6

Aim: Write a program to show all the operations of a stack.

Theory:

A stack is a data structure which is used to store data in a linear fashion. Stack follows a Last in, First out (LIFO) principle i.e. the data which is inserted most recently in the stack will be deleted first from the stack. There are many stack operations that can be performed on a stack. Some of them are push(), pop(), peek(), isEmpty(), isFull(), size().

Stack operations are generally used to extract information and data from a stack data structure.

1. push(): Push is a function in stack definition which is used to insert data at the stack's top.
2. pop(): Pop is a function in the stack definition which is used to remove data from the stack's top.
3. topElement() / peek(): TopElement / Peek is a function in the stack which is used to extract the element present at the stack top.
4. isEmpty(): isEmpty is a boolean function in stack definition which is used to check whether the stack is empty or not. It returns true if the stack is empty. Otherwise, it returns false.
5. isFull(): isFull is a function which is used to check whether the stack has reached its maximum limit of insertion of data or not
6. size(): Size is a function in stack definition which is used to find out the number of elements that are present inside the stack.

Source Code:

```
#include <iostream>

using namespace std;

const int MAX = 5;

class Stack {
   int arr[MAX];
   int top;
public:
   Stack() {
      top = -1;
```

```
}
bool isFull() {
   return top == MAX - 1;
}
bool isEmpty() {
   return top == -1;
}
void push(int x) {
   if (isFull()) {
      cout << "Stack overflow\n";
      return;
   }
   top++;
   arr[top] = x;
   cout << x << " inserted\n";
}
void pop() {
   if (isEmpty()) {
      cout << "Stack underflow\n";
      return;
   }
   cout << arr[top] << " deleted\n";
   top--;
}
int peek() {
   if (isEmpty()) {
      cout << "Stack is empty\n";
      return -1;
```

```
        }
        return arr[top];
    }
    int size() {
        return top + 1;
    }
    void display() {
        if (isEmpty()) {
            cout << "Stack is empty\n";
            return;
        }
        cout << "Stack elements: ";
        for (int i = top; i >= 0; i--) {
            cout << arr[i] << " ";
        }
        cout << "\n";
    }
};
int main() {
    cout << "Deepayan Chakravarti" <<endl;
    cout << "04714802724" <<endl;
    Stack s;
    cout << (s.isEmpty() ? "true" : "false") << "\n";
    cout << (s.isFull() ? "true" : "false") << "\n\n";

    s.push(10);
    s.push(20);
    s.push(30);
```

```
    s.push(40);
    s.push(50);
    s.push(60);
    cout << (s.isFull() ? "true" : "false") << "\n";
    cout << (s.isEmpty() ? "true" : "false") << "\n";
    cout << "Current stack size: " << s.size() << "\n";
    cout << "Top element (peek): " << s.peek() << "\n\n";
    s.display();
    s.pop();
    s.pop();
    s.display();
    cout << "New top element: " << s.peek() << "\n";
    cout << "New stack size: " << s.size() << "\n\n";
    s.pop();
    s.pop();
    s.pop();
    s.pop();
    cout << (s.isEmpty() ? "true" : "false") << "\n";
    return 0;
}
```

Output:


Problems
Output
Debug Console
Terminal
Ports
Code
PS C:\Users\Deepayan\Documents\MAVERICK\labs\5th-sem\cd-lab> cd "c:\Users\Deepayan\Documents\MAVERICK\labs\5th-sem\cd-lab\" ; if ($?
-o exp6 } ; if ($?) { .\exp6 }
Deepayan Chakravarti
04714802724
true
false
10 inserted
20 inserted
30 inserted
40 inserted
50 inserted
Stack overflow
true
false
Current stack size: 5
Top element (peek): 50
Stack elements: 50 40 30 20 10
50 deleted
40 deleted
Stack elements: 30 20 10
New top element: 30
New stack size: 3
30 deleted
20 deleted
10 deleted
Stack underflow
true
PS C:\Users\Deepayan\Documents\MAVERICK\labs\5th-sem\cd-lab>